

Adding Even More Theoretical Precision

To get even more precise, we should try to reason carefully about the steps required for insertion sort. Let's imagine, once again, that we have a new unsorted list with n elements. In [Table 4-2](#), we proceed through each step of insertion sort individually and count the steps.

Table 4-2: *Counting the Steps in Insertion Sort*

Description of actions	Number of steps required for pulling the file from the old cabinet	Maximum number of steps required for comparing to other files	Number of steps required for inserting the file into the new cabinet
Take the first file from the old cabinet and insert it into the (empty) new cabinet.	1	0. (There are no files to compare to.)	1
Take the second file from the old cabinet and insert it into the new cabinet (that now contains one file).	1	1. (There's one file to compare to and we have to compare it.)	1
Take the third file from the old cabinet and insert it into the new cabinet (that now contains two files).	1	2 or fewer. (There are two files and we have to compare between 1 of them and all of them.)	1
Take the fourth file from the old cabinet and insert it into the new cabinet (that now contains three files).	1	3 or fewer. (There are three files and we have to compare between 1 of them and all of them.)	1
...	...	...	...
Take the n th file from the old cabinet and insert it into the new cabinet (that contains $n-1$ files).	1	$n-1$ or fewer. (There are $n-1$ files and we have to compare between one of them and all of them.)	1

If we add up all the steps described in this table, we get the following maximum total steps:

- Steps required for pulling files: n (1 step for pulling each of n files)
- Steps required for comparison: up to $1 + 2 + \dots + (n-1)$